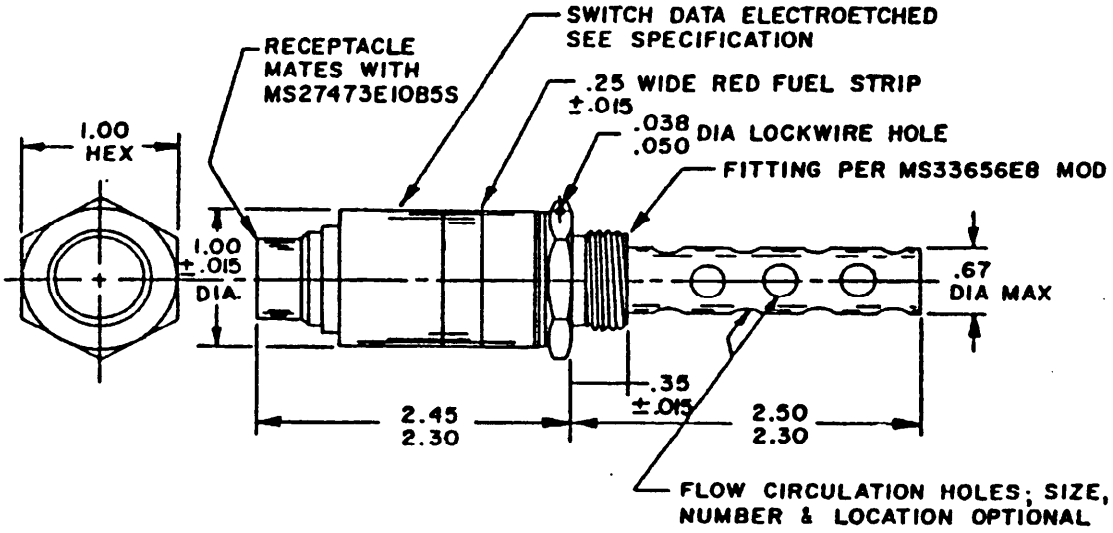
MILITARY SPECIFICATION SHEET

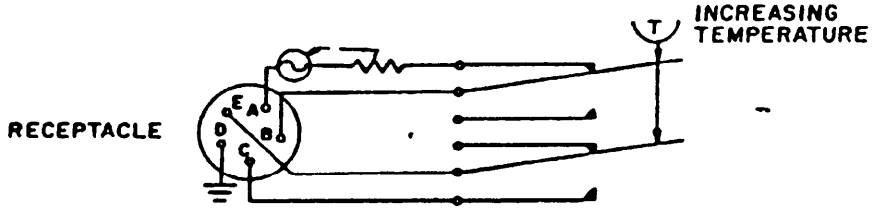
SWITCH, THERMOSTATIC (VOLATILE LIQUID), HERMETICALLY SEALED, HEATING UNIT ACTUATED, (SPST) NORMALLY OPEN

This specification is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-S-28827.



Inches	mm
.015	0.38
.038	0.97
.050	1.27
.25	6.4
.35	8.9
.67	17.0
1.00	25.4
2.30	58.4
2.45	62.2
2.50	63.5



ELECTRICAL SCHEMATIC
AT ROOM AMBIENT

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for gereral information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5930

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REQUIREMENTS

Dimensions, configurations, and electrical schematic: See figure 1.

Weight: .40 pound.

Operating point tolerances: See tables III, IV, or V as applicable.

Proof pressure: 120 psig.

Proof temperature: 100°F above the maximum increasing set point.

Burst pressure: 180 psig.

Response time: Heater, 0.15 second per degree of F setting.
Probe, 30 second maximum for any setting.

Electrical ratings:

Operating voltage: 28 volts ±5 V dc.

Current rating: .5 ampere resistive.

Electrical requirement, heater:

Voltage	28 V dc
Power	75 watts maximum
Current	2.7 amperes maximum

Seal: Electrical chamber hermetic.

Media: JP4 and JP5 or any media compatible with wetted materials.

High temperature (ambient operating and nonoperating): +365°F maximum.

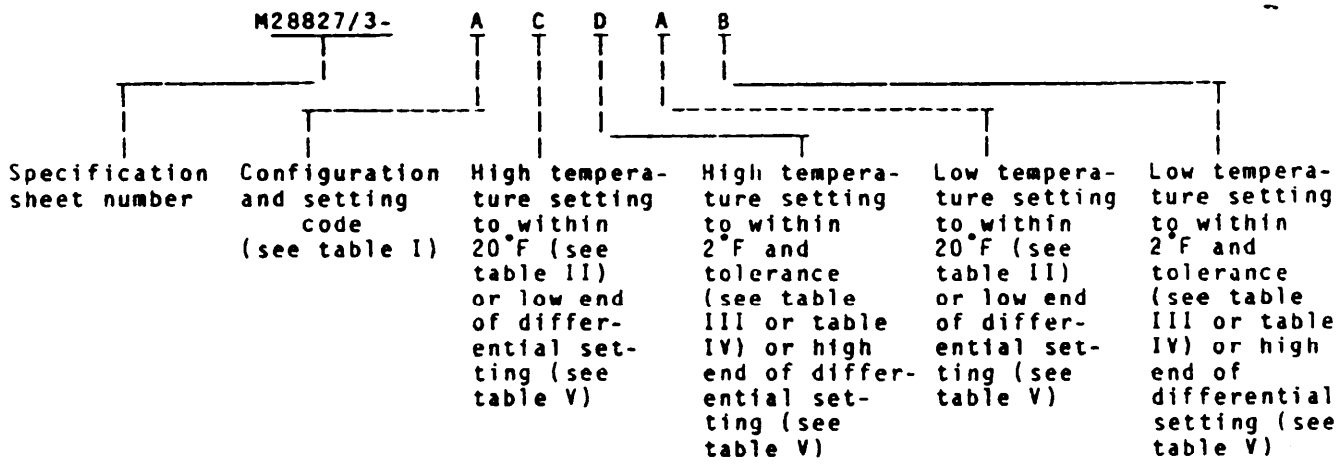
Low temperature (ambient operating and nonoperating): -65°F.

Altitude: 80,000 feet.

Endurance: 25,000 cycles.

Materials: All wetted materials shall be 300 series stainless steel.

Part number shall be assigned as illustrated below:



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Example of part numbers:

M28827/3-ACDAB identifies a switch which actuates on increasing temperatures at 146°F ±5°F and deactuates at 102°F ±5°F

M28827/3-CDBEH identifies a switch that actuates at 162°F±5°F with a 20°F/35°F differential temperature.

M28827/3-EBCDE identifies a switch which actuates by 124°F maximum and deactuates by 168°F maximum.

TABLE I. Code characters for combinations of configurations and temperature setting modes.

High setting	Low setting	Configuration
At 1/	At 1/	A
At 1/	By 2/	B
At 1/	Differential 3/	C
By 2/	At 1/	D
By 2/	By 2/	E
Differential 3/	At 1/	F

1/ Setting values are designated by characters from tables II and III.

2/ Setting values are designated by characters from tables II and IV.

3/ Setting values are designated by characters from table V.

TABLE II. Temperature setting to within 20°F.

Temperature	Code	Temperature	Code
100	A	260	J
120	B	280	K
140	C	300	L
160	D	320	M
180	E	340	N
200	F	360	P
220	G		Q
240	H		R

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TABLE III. Temperature setting within 2°F and tolerance.

Unit	Tolerance		
	±5	±10	±25
0	A	L	W
2	B	M	X
4	C	N	Y
6	D	P	Z
8	E	Q	1
10	F	R	2
12	G	S	3
14	H	T	4
16	J	U	5
18	K	V	6

TABLE IV. Temperature setting to within 2°F.

Unit	Maximum or minimum
0	A
2	B
4	C
6	D
8	E
10	F
12	G
14	H
16	J
18	K

TABLE V. Code letters for differential temperature settings.

Code letter	Differential temperature (°F)
A	0
B	5
C	10
D	15
E	20
F	25
G	30
H	35
J	40
K	45
L	50
M	60
N	75

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Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5930-N583-2)

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